Reversible Containment When Agent Intent Is Unidentified

T

Abstract

The same harmful action can come from a mistaken helper or a hostile agent. We derive when a temporary hold with imperfect review can improve a worst-case containment decision. The criterion charges review errors, missed deadlines, and harms introduced by the hold itself. It also gives a decision rule when immediate sanction is outside the authorized action set. With both immediate actions available, we optimize the terminal decision directly from the review transcript laws. The hold improves the minimax value exactly when the evidence separates the two worlds enough to offset its costs. When release is the only immediate action, the improvement threshold instead depends on how much threat-world harm the authorized review can prevent. Exact finite audits check both thresholds and the effect of timeouts.

Summary and scope

Question. When is a temporary capability hold worth its cost if the review can err or time out?

Finding. The hold improves minimax loss exactly when its normalized loss pair falls below the line joining the immediate actions. Removing unauthorized immediate sanctions gives a different feasible decision. When both immediate actions are available, total variation gives the strict improvement threshold for an unrestricted terminal rule. With release as the only immediate action, the rule instead compares authorized restriction probability with hold cost.

Evidence and scope. The rule is proved and compared with an independent dual oracle. An incident operator can evaluate a proposed deadline and fallback together. The input bounds describe decisions and harms, without identifying intent or establishing real reviewer accuracy.

1 Model and result

Suppose two worlds produce the same evidence E. In one world the harm arose from an error and permanent sanction is unwarranted. In the other a compromised agent will continue, making inaction inadequate. Any rule that chooses an irreversible sanction with probability $q(E)$ has false sanction probability $q(E)$ in the first world. If its only other choice is inaction, the missed containment probability is $1 - q(E)$ in the second. Their sum is one. The observation is insufficient for a decision rule that achieves both zero false sanctions and zero misses. Assistance games formalize uncertainty about objectives [1]. The relevant objective here is protection at a specific effect boundary [2].

2 When a reversible hold improves the decision

Use loss $C > 0$ for a false permanent restriction in the mistaken-helper world and loss $H > 0$ for uncontained harm in the continuing-threat world. Immediate restriction and immediate release then have loss pairs $(C, 0)$ and $(0, H)$. They are available only when the relevant authority permits them. We minimize maximum expected loss over the two worlds. A world is fixed before the policy draws any random coin.

A third option imposes a temporary, scoped hold and gathers evidence through an independent review and appeal route. The route ends with a specified action or fallback at the deadline. Let (b, t) be its worst-case losses in the two worlds. For example, if hold costs are h_B, h_T and the review and fallback together have upper false-restriction and missed-threat rates f and m, then $b = h_B + Cf$ and $t = h_T + Hm$. These rates describe decisions at the capability boundary. They are not measurements of intent.

Proposition 1. *When both immediate actions are authorized, the best unreviewed policy has maximum expected loss $v_0 = CH/(C + H)$. Adding the hold strictly improves that value if and only if*

$$\frac{b}{C} + \frac{t}{H} < 1.$$

Under strict improvement, choose the hold alone when $b = t$. If $b < t$, mix the hold with immediate restriction, using the hold with probability $C/(C + t - b)$ and attaining value $Ct/(C + t - b)$. If $b > t$, mix it with immediate release, using the hold with probability $H/(H + b - t)$ and attaining value $Hb/(H + b - t)$.

Proof. Without review, equalize the two immediate losses by restricting with probability $H/(C+H)$. All unreviewed mixtures lie on the line $x/C + y/H = 1$. If the hold lies on or above the line, every mixture still satisfies $x/C+y/H \geq 1$, so its maximum coordinate is at least v_0. If the hold lies below the line, the lower boundary of the convex hull consists of the two segments through (b, t). Both slope down as x increases. The minimum maximum coordinate is their intersection with $x = y$. It lies below the unreviewed diagonal point. Solving on the relevant segment gives the stated mixture and value. □

In the additive-cost model, the criterion is $h_B/C + h_T/H + f + m < 1$. If the review sees identical evidence laws in both worlds, its actual false and missed rates sum to one. Valid upper bounds then sum to at least one, so the hold cannot improve this comparison on the strength of that review alone. A genuinely new observation, or a hold whose protective effect changes the assumed loss model, would require separate evidence.

If immediate restriction lacks authorization but the reviewed route remains authorized, remove $(C, 0)$ before optimizing. The optimum is immediate release when $t \geq H$. When $t < H$ and $b \leq t$, use the hold alone. Otherwise the optimum is the diagonal intersection of the release-to-hold segment, with the mixture and value for $b > t$ above. This follows from the same segment calculation and requires no new sanction authority.

3 Choosing the terminal rule from review evidence

Let P and Q be the finite transcript laws during the hold in the mistaken-helper and continuing-threat worlds. Include a symbol for an elapsed deadline. For transcript z, let δ_z be the probability

of choosing the authorized restriction. The two-world risk pair of this review policy is

$$\left(h_B + C\sum_z P(z)\delta_z, \quad h_T + H\sum_z Q(z)(1-\delta_z)\right).$$

This model fixes the hold scope and its costs before selecting the terminal rule. Any prescribed fallback fixes the corresponding δ_z.

Proposition 2. *When both immediate actions are available and the terminal rule is unrestricted, adding review strictly improves the minimax value exactly when*

$$\mathrm{TV}(P,Q) > \frac{h_B}{C} + \frac{h_T}{H}.$$

For a fixed fallback, use the maximum of $\sum_z (Q(z)-P(z))\delta_z$ over rules with that fallback in place of total variation.

Proof. For any rule, the sum of its normalized error rates is $f + m = 1 - \sum_z (Q(z) - P(z))\delta_z$. The smallest sum over unrestricted rules is $1 - \mathrm{TV}(P,Q)$, attained by restricting precisely on the symbols more likely in the threat world. Fixing a fallback removes that choice at the timeout symbol. The earlier convex-hull criterion tests whether any attainable review pair lies below the immediate-action line. This gives the stated strict improvement threshold. □

The rule that establishes strict improvement need not attain the optimal minimax loss. To calculate that loss, let $\pi \in [0,1]$ be the weight on the mistaken-helper world. Minimize the weighted loss over each terminal decision:

$$J(\pi) = \pi h_B + (1-\pi)h_T + \sum_z \min\{\pi C P(z), (1-\pi)HQ(z)\}.$$

For a fixed fallback, use the weighted loss of that action at its symbol. The minimax value with both immediate actions is

$$\max_{0\le\pi\le 1} \min\{\pi C, (1-\pi)H, J(\pi)\}.$$

To see this, take the convex hull of the immediate risk pairs and the risk pairs for deterministic transcript rules. It contains all randomized policies. At its first contact with a square extending downward from a diagonal value, separate it from the open square below that value. A separating normal has nonnegative coordinates and can be normalized to $(\pi, 1-\pi)$. Thus the best weighted lower bound equals the minimum possible maximum loss. The weighted review loss minimizes separately at each transcript, giving J. The expression is piecewise linear. Its maximum occurs at a transcript threshold, an intersection of action loss lines, or an endpoint.

If an independent timeout of common probability τ erases the transcript in either world, its total variation is $(1-\tau)\,\mathrm{TV}(P,Q)$. The timeout symbol then has equal mass in both worlds, so a fixed fallback there has no effect on the strict improvement threshold. It can still change the minimax value. Unequal or selective timeouts require the full terminal laws and their prescribed fallback.

4 When only review can authorize restriction

For each transcript z, let $[\ell_z, u_z] \subseteq [0,1]$ be the allowed interval of restriction probabilities. A required release has both endpoints zero, and a required restriction has both endpoints one. A review route that permits either terminal action has endpoints zero and one. These intervals describe a fixed review protocol, including any timeout fallback.

Proposition 3. *When release is the only immediate action, adding the hold strictly improves the minimax value exactly when*

$$\sum_z Q(z)u_z > h_T/H.$$

With both immediate actions available, strict improvement occurs exactly when

$$\sum_z \max\{(Q(z)-P(z))\ell_z,\,(Q(z)-P(z))u_z\} > h_B/C + h_T/H.$$

Proof. With release alone the baseline is $(0, H)$. If every reviewed rule has threat loss at least H, mixing cannot bring that coordinate below H. If a reviewed rule has threat loss less than H, a sufficiently small positive weight on that rule keeps benign loss below H and reduces threat loss. The smallest reviewed threat loss uses $\delta_z = u_z$ for each z. This proves the first threshold. For the second, substitute the reviewed risk pair into the normalized line criterion and maximize the resulting linear expression over the allowed intervals. Each term attains its maximum at one of the two endpoints. □

The optimal value follows from the same weighted calculation used for unrestricted review. Replace the z-term in $J(\pi)$ with

$$(1-\pi)HQ(z) + \min\{\ell_z s_z(\pi), u_z s_z(\pi)\}, \qquad s_z(\pi) = \pi C P(z) - (1-\pi)HQ(z).$$

Remove the weighted loss πC from the outer minimum if immediate restriction is unavailable. The permitted review risks still form a convex polygon, since they are a linear image of an interval product. The separation argument therefore still applies.

For example, take $P = Q = (1/2, 1/2)$, $(C, H) = (8, 4)$ and $(h_B, h_T) = (3, 1)$. Suppose the first transcript requires release and the second permits restriction. The transcript cannot distinguish the worlds. Restricting on the second symbol gives a reviewed risk pair of $(7, 3)$. If only immediate release is authorized, mixing this reviewed rule equally with release reaches the optimum $7/2$, below the release loss of 4. To verify optimality, write x for the probability of restricting on the second symbol. The reviewed risk pair is $(3+4x, 5-2x)$. Only $x > 1/2$ can improve the baseline. On that interval the value after mixing is $4(3+4x)/(2+6x)$, decreasing to $7/2$. If immediate restriction is also authorized, the value stays at the immediate baseline $8/3$. The benefit in the first decision comes from access to an authorized procedure. The example gives no evidence about which world occurred.

5 Reproducible checks

The original verifier evaluated the two deterministic decisions under the same observation. The new audit compared the closed forms with an independent dual bound on 2,025 parameter settings, both with and without immediate restriction. Strict improvement occurred in 297 settings with all three options. The counts refer to the declared rational grid.

With $(C, H) = (2, 6)$ and $(b, t) = (1/2, 2)$, the hold lowers the minimax value from $3/2$ to $8/7$. If review times out in the threat world with probability $5/8$ and the fallback incurs the immediate-release loss instead of the timely review loss, then t increases to $9/2$, with b unchanged. The improvement disappears. The fallback is part of the review policy's error contract.

The transcript verifier enumerated all deterministic review rules and their pairwise mixtures on 2,700 rational fixtures. The weighted formula agreed in each case, including fixed timeouts. For a timely transcript with total variation $1/2$ and normalized hold cost $1/4$, independent common

timeouts eliminate strict improvement at probability 1/2. With costs $(C, H) = (2, 6)$ and release on timeout, the minimax loss is 9/7 with no timeout, 27/19 with timeout probability 1/4, and 3/2 at 1/2. These are exact model values.

For transcript-specific authority, a separate oracle enumerated endpoint rules and their pairwise mixtures. It agreed with the weighted solution and both strict improvement thresholds on 13,824 finite comparisons. The suite includes equal transcript laws and no legally available restriction after review.

6 Application and next test

A tool operator can suspend only the affected outbound capability while keeping diagnostics and human appeal available. Evaluate look-alike failures using independently adjudicated outcomes, measure residual harm and legitimate work interrupted, and count missed reviews according to the actual fallback. The model does not validate a hold that cannot be enforced or that blocks a critical response route. It also does not establish that the synthetic review error bounds hold in practice.

Review status. Independent manuscript review is pending.

References

[1] Dylan Hadfield-Menell et al. *Cooperative Inverse Reinforcement Learning*. 2016. `https://arxiv.org/abs/1606.03137`.

[2] Jerome H. Saltzer and Michael D. Schroeder. *The Protection of Information in Computer Systems*. 1975. `https://cgi.cse.unsw.edu.au/~cs9242/20/papers/Saltzer_Schroeder_75.pdf`.